

UTXO: A Belated DNS White Paper

From 13 Years of HTB Payment Addressing Sedimentation to the Paradigm Shift of BIS Unified Ledger

Version: 2.0 | **Date:** July 2026 | **Author:** J. Tian (uw2icg-core)

Abstract: UTXO-DNS is not a concept that emerged from nowhere. It originates from the intellectual sedimentation of the 2013 HTB (HIEP: HTB Internet E-wallet Protocol) payment addressing draft — embedding payment endpoint identifiers into DNS. Thirteen years later, when stablecoins have reached an annual settlement volume of \$28 trillion, BlackRock has initiated RWA tokenization with \$13 trillion in assets under management, and the BIS Unified Ledger has completed cross-border atomic settlement verification involving 7 central banks, the gap in DNS's payment naming layer can no longer be ignored. introduction of blockchain technology, but the filling of a never-before-filled void: unified naming for payment endpoints.

I. Introduction: DNS's 50-Year "Blind Spot" and HTB's 13 Years of Sedimentation

DNS was born in 1983. Its design goal was clear and singular: to map human-readable domain names to machine-readable IP addresses. But in 2026, a "payment endpoint" is no longer just an IP address. It is a set of information:

Question	Can DNS Answer?
Who owns this domain? (Identity)	 No
Is the transaction compliant? (Compliance)	 No
Where should the payment be sent? (Payment Address)	 No
Does settlement have finality? (Finality)	 No
Whose data? Who has rights?	 No



DNS's "blind spot" lies in the fact that it resolves only the "where" question, yet cannot resolve "who," "what rules," "how to pay," "how to settle," or "whose data."

In 2013, the HTB (HIEP: HTB Internet E-wallet Protocol) draft was submitted to the IETF — embedding payment addressing into DNS. This was a judgment made 13 years ahead of its time: payment endpoints require a unified naming layer. At the time, it was overlooked due to lack of application scenarios.

In June 2026, UTXO-DNS received an official response from IETF DNSOP, with a proposal to present a 10-minute technical talk at the IETF DNSOP Vienna 126 global conference of 1,000 technical participants, under the direct guidance of IESG member Eric Vyncke's IPv6 extension tools 【1+L2-L3】. Thirteen years — from being overlooked to being invited — the DNS community has finally recognized that the payment naming layer falls within its purview.

UTXO-DNS is not a concept that emerged from nowhere. It is the inheritance and upgrade of HTB's ideas after 13 years of technical evolution: from HTTP protocol payment identifiers to DNS protocol UTXO RR type; from simple domain name-to-bank-card binding to VRF cryptographic verification + vLEI legal entity identity + PRN penetrative regulation【1+L3】. During these 13 years, it was continuously refined through multiple discussions at the W3C Web Payments Paris Conference, Beihang University Roundtable, with contributions from former W3C Chair Jeff Jaffe.

II. The Fragmentation of Payment Naming — and History Repeating Itself in the AI Era

2.1 Payment Fragmentation: Duplicate Database Construction

When HTB was proposed in 2013, the core judgment was: if DNS could unify payment endpoint naming, there would be no need for each payment system to build its own "address book" database.

Thirteen years later, this judgment has been fully validated:

Payment

System

Address Book Format

Interoperability

Alipay

Phone number/Email



WeChat Pay

WeChat ID/Phone number



PayPal

Email address



SWIFT

IBAN + BIC



USDC/USDT

0x... address



Every payment system is repeatedly building its own address book database, isolated from one another, resulting in enormous resource waste and user burden.



2.2 AI Era Fragmentation: History Accelerating

What is alarming is that the AI era is repeating the same fragmentation at an accelerated pace:

AI Service Provider	User Data Storage	User Identity Identifier	Interoperability
ChatGPT	Conversation history	Email/Phone	
	locked in OpenAI	number	
Claude	Conversation history	Email/Phone	
	locked in Anthropic	number	
DeepSeek	Conversation history	Email/Phone	
	locked in DeepSeek	number	
Various AI apps	Independent storage	Independent accounts	

Users' "digital selves" are being fragmented — your conversation data, your knowledge contributions, your intent history are scattered across dozens of mutually isolated silos. You cannot manage them uniformly, cannot migrate them, cannot control how they are used, and cannot receive the rewards you deserve from them 【1†L4-L5】.

2.3 One White Paper Addressing Two Fragmentations

Simultaneously

The unique aspect of UTXO-DNS is that it simultaneously addresses payment fragmentation and AI fragmentation — because both are essentially the same problem: the internet lacks a "unified naming layer for personal digital identity" 【1†L5】.

Fragmentation

Type	Root Problem	UTXO-DNS Response
Payment	Each payment system	.utxo as unified payment



Fragmentation

Type

Root Problem

UTXO-DNS Response

Fragmentation	maintains its own address book	endpoint naming
AI Data	Each AI service provider	.utxo as unified data ownership
Fragmentation	owns user data Compute resources	identity
Compute	cannot be uniformly	.utxo as unified identifier for
Fragmentation	addressed and configured	compute RWA

Payment, data, and compute — all three will be uniformly named by *.utxo* 【1+L6】.

III. UTXO-DNS: DNS Protocol Extension Based on IETF Standards

3.1 Technical Architecture: Fully Built on Existing IETF Standards

UTXO-DNS is fully built on existing IETF standards:

- RFC 1034/1035 — DNS Framework
- RFC 6891 — EDNS0 Extensions
- RFC 7217 — Stable Privacy Addressing Paradigm
- RFC 9381 — VRF (Verifiable Random Function)
- RFC 8949 — CBOR Encoding

Core Innovation: Extending the parameter source of the IPv6 Interface Identifier (IID) generation function in RFC 7217 from "device-local MAC address" to "UTXO transaction credentials" 【1+L6-L7】.



3.2 New DNS RR Type: UTXO RR (Code 260)

The RDATA of UTXO RR contains three core parts:

1. **Version and Flags** — supporting future extensions and algorithm upgrades (including post-quantum cryptography migration paths)
2. **CBOR-encoded Endpoint Identifier** — capable of carrying identity credentials, compliance policy references, and blockchain addresses
3. **Optional VRF Proof** — providing cryptographic-level offline verification capability

This is a **universal container** — not bound to any specific implementation. The identity field can carry GLEIF vLEI or eIDAS; the payment endpoint can be a blockchain address or a traditional bank account identifier 【1†L7】.

3.3 Full Compatibility with Existing DNS

EDNS0 Option Code is used for capability discovery. Per RFC 6891, unknown EDNS0 options are ignored by resolvers, and queries do not fail. UTXO RR is a Data RR, not a Meta-RR 【1†L7】.

IV. Global Evidence: Irreversible Underlying Drivers

4.1 Stablecoins: \$75 Billion Supporting \$28 Trillion Settlement

In 2025, the global mainstream stablecoin annual transaction volume reached \$28 trillion, exceeding the combined annual transaction scale of Visa and Mastercard 【1†L8】.

Metric	Data
USDC Circulation (end of 2025)	\$75.3 billion (72% YoY growth)
Stablecoin Annual Settlement	\$28 trillion
Cross-border Remittance Cost	Reduced from 6.62% to near zero

\$28 trillion means stablecoins have formed a global settlement network parallel to SWIFT, without relying on physical ATMs, bank branches, or correspondent banking relationships. The pain points that plague global travelers — high ATM failure rates, inconvenient cross-border cash withdrawals, and exchange rate losses — are exactly the real-world scenarios that stablecoin payments need to address.

But the core bottleneck facing stablecoins is: payment endpoints lack unified naming. Users need to remember 0x... addresses rather than human-readable domain names. UTXO-DNS is precisely filling this gap 【1†L8】.



4.2 BlackRock BUIDL: The Vanguard of \$13 Trillion RWA

Tokenization

BlackRock's total assets under management reached \$13.46 trillion in Q3 2025, up 17% year-over-year 【1+L8】. Its tokenized money market fund BUIDL, launched in March 2024, has exceeded \$2.8 billion in scale and expanded to 8 chains 【1+L8】.

Timeline	Event	Strategic Implication
Mar 2024	BUIDL launches on Ethereum	Testing RWA tokenization
Nov 2025	Expands to BNB Chain, Solana, and 6 other chains	Multi-chain strategy; 68% of AUM deployed outside Ethereum
Nov 2025	Integrated with Binance as institutional collateral	Bridging CeFi-DeFi liquidity
Nov 2025	Integrated with Uniswap, Euler, and other DeFi protocols	BUIDL can serve as lending collateral

BlackRock CEO Larry Fink explicitly stated: "Tokenization could be the single most important factor in the evolution of the global financial system." If all \$5.3 trillion of ETFs were tokenized and placed in digital wallets, investors could seamlessly and costlessly convert between cash, stocks, and bonds without intermediaries.

BUIDL's success is not because institutions want "blockchain exposure" — but because it solves specific operational problems: 24/7 liquidity, T+0 settlement, and direct usability as collateral. Traditional money market funds settle at T+1; BUIDL compresses this to near-instantaneous.

BlackRock's strategy is clear: first validate RWA tokenization with BUIDL, then gradually move \$13 trillion in assets onto the chain 【1+L8-L9】.

4.3 RWA Market Forecast: \$13 Trillion by 2030

Institution	Forecast	Timeline
BlackRock	Tokenized asset market reaches \$13 trillion	2030



Institution	Forecast	Timeline
Deloitte	Real estate RWA token market reaches \$4 trillion	2035
Boston Consulting Group	Global tokenized asset market reaches \$16 trillion	2030

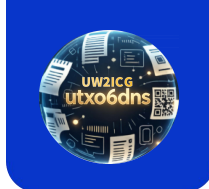
RWA tokenization is moving from proof-of-concept to large-scale deployment 【1†L9】.

V. Hong Kong Stablecoin Legislation: The Institutional Cornerstone of the RWA Trend

5.1 Establishment of a Compliance Framework

On August 1, 2025, Hong Kong's Stablecoins Ordinance (Cap. 656) came into effect, establishing a mandatory licensing regime for fiat-referenced stablecoins (FRS) 【1†L4-L5】.

Timeline	Event	Significance
Aug 1, 2025	Stablecoins Ordinance takes effect	World's first comprehensive stablecoin regulatory framework
Apr 10, 2026	First licenses issued (HSBC, Matrixport)	Compliant stablecoins officially launched



Timeline	Event	Significance
2026-	Stablecoins	Payment infrastructure for RWA
	officially issued	tokenization

5.2 Hong Kong's Positioning in the RWA Trend

The Hong Kong Monetary Authority's Policy Statement 2.0 clarifies the institutional path for RWA tokenization:

- **Project Ensemble:** Regulatory sandbox for financial asset tokenization
- **EnsembleTX Platform:** Testing ground for tokenized asset trading
- **Tokenized Deposits:** Providing settlement medium for RWA transactions

Hong Kong's stablecoins are accelerating RWA trend development — compliant stablecoins provide fiat on/off-ramps and settlement media for RWA tokenization 【1+L5】 .

5.3 Institutional Barriers to RWA Tokenization Are Being Eliminated

The core bottlenecks of RWA tokenization are being overcome one by one 【1+L5-L6】 :

Bottleneck	Solution	Status
Compliance	Hong Kong stablecoin legislation +	Launched
Fragmentation	PRN multilateral regulation	
Liquidity	CoCaDEX unified trading layer	In progress
Dispersion		
Identity	vLEI + PRN nodes	Integrated
Verification Gap		
Fiat On/Off-Ramp	Compliant stablecoins (HKDAP, etc.)	Licensed



Bottleneck	Solution	Status
Cross-border Settlement	Digital RMB + multilateral central bank bridge	In progress

The institutional infrastructure for RWA tokenization is taking shape.

VI. CoCaDEX: Digital Wall Street — AI Compute Resource Allocation and RWA Globalization

6.1 CoCaDEX's Strategic Positioning: The Wall Street of the Digital Age

If traditional Wall Street is the capital allocation center of the industrial age — channeling capital from savers to producers through stocks, bonds, and derivatives — then CoCaDEX is the capital allocation center of the digital age, but with capabilities far exceeding traditional Wall Street 【1+L9】.

Wall Street	Traditional Wall Street	CoCaDEX (Digital Wall Street)
Function		
Capital Formation	Corporate stocks/IPOs	RWA asset tokenization issuance (compute/real estate/infrastructure)
Asset Trading	Stocks/Bonds/Derivatives	RWA tokens + Data UTXO + Compute
Price Discovery	Exchange quotes	Fair value oracle (AI-driven pricing)



Wall Street

	Traditional Wall Street	CoCaDEX (Digital Wall Street)
Function		
Risk	Hedging/Insurance	Circuit breakers + Insurance pools
Management	products	+ Perpetual protocols
Compliance	SEC/FINRA regulation	PRN nodes + HKMA licensed + Multilateral regulation
Inclusivity	High barrier for accredited investors	Fragmented investment, globally accessible

CoCaDEX's uniqueness lies in the fact that it does not merely trade financial assets — it trades the factors of production in the AI era: compute, data, and model capabilities 【1+L9-L10】.

6.2 CoCaDEX — AI Compute Resource Allocation Center

The core contradiction of the AI era: compute resources are highly concentrated, while demand is extremely decentralized 【1+L11】.

Current Compute Market	CoCaDEX Compute Allocation Center Solution
NVIDIA/Microsoft/AWS oligopoly	Compute RWA tokenization — any GPU owner can issue compute tokens
Compute prices unilaterally set by giants	Market matching + fair value oracle — supply and demand determine price
Small teams/individuals struggle to access compute	Fragmented leasing — by the hour, by precision



CoCaDEX Compute Allocation Center Solution	
Current Compute Market	
Compute idleness and shortage coexist	Global compute matching market — idle compute automatically flows to demand
AI service providers independently address	Unified <code>.utxo</code> naming — all AI services accessible via domain names

CoCaDEX is not a "crypto exchange," but a global allocation market for AI-era compute, data, and model capabilities 【1+L12】.

6.3 Data Rights Confirmation: Returning Data to Users, Making AI a Renter

The fundamental flaw of the current AI model:

Users contribute data through AI conversations; this data is used to train models; models become smarter; service providers charge higher fees — but users receive no compensation and cannot even control their own conversation data 【1+L15】.

CoCaDEX's Data UTXO Model 【1+L15-L16】:

Current User Data Status	CoCaDEX Data UTXO Model
Data is freely obtained by AI service providers	Users have full ownership of data (bound to <code>.utxo</code> identity)
Data cannot be traded	Data UTXO can be authorized, priced, and traded on CoCaDEX
Data value cannot be quantified	Fair value oracle prices data (based on quality + scarcity + domain)
Data usage cannot be	On-chain UTXO records complete authorization



Current User Data Status

traced

Users receive no benefit

CoCaDEX Data UTXO Model

chain — who used it, when, for what purpose

Users earn revenue: when conversation data is used, 70% of rental fee goes to the user

Data sovereignty moves from servers to user wallets — this is the most profound transformation of CoCaDEX as Digital Wall Street 【1†L16】.

6.4 Inclusive Finance: Enabling Global Private Real Estate to

Generate Cash Flow

Against the macroeconomic backdrop of BlackRock — with \$13 trillion in global assets under management — driving its global RWA strategy, the economic structure will inevitably undergo a new round of adaptive adjustments, which will further transmit to the tokenization process in the monetary policy domain. Under this framework, millions of households will be able to convert their property equity into tradable digital assets, and can rely on the BUIDL fund mechanism to achieve dual-income accumulation 【1†L17】.

The Path Provided by CoCaDEX 【1†L17】:

- Every property obtains a *.utxo* domain → binding property title hash
- Generate RWA tokens → fragmentation of rental income rights
- Global capital can invest simultaneously in property rental income and BUIDL funds through CoCaDEX, enjoying double-layer compound returns

Property owners gain additional cash flow → alleviating living pressure → stimulating consumption markets

All of this requires no government investment — achievable solely through UTXO-DNS's naming layer and CoCaDEX's market layer 【1†L17】.

VII. *.utxo* Integration with BIS Unified Ledger: Paradigm Shift in Central Bank-Level Functionality

7.1 Core Breakthrough of BIS Unified Ledger

In May 2026, BIS released the Project Agora final report. Participants included the Federal Reserve Bank of New York, the Bank of England, the Bank of Japan, and 4 other major central banks, along with over 40 private financial institutions. The core conclusion: tokenized central bank reserve funds can achieve second-level atomic settlement for cross-border payments【1†L17】.



Core Capability	Description
Atomic Settlement	"All-or-nothing" settlement across currencies and jurisdictions, eliminating settlement risk
Programmable Compliance	AML/CFT screening and sanctions checks executed in parallel via smart contracts
Layered Architecture	Each central bank retains full autonomy over its own currency and operations on the shared ledger
Settlement Finality	Legally binding settlement finality achievable across all participating jurisdictions

7.2 *.utxo*'s Core Role in the BIS Unified Ledger

The BIS Unified Ledger solves the "settlement" problem — enabling atomic settlement of tokenized assets on a shared platform. But it faces a fundamental problem: when central banks from different jurisdictions, different commercial banks, and different asset types all interact on the same ledger, how can these participants and assets be uniformly identified, discovered, and addressed 【1†L18】 ?

.utxo provides exactly this layer — the "naming layer" of the unified ledger 【1†L18】 :

Unified Ledger Participant	Traditional Identification	<i>.utxo</i> Unified Naming
Central Bank	Independent identification systems of each central bank	<i>fed.utxo</i> , <i>boe.utxo</i> , <i>boj.utxo</i>
Tokenized Reserves		
Commercial Bank	BIC + Account Number +	<i>jpmorgan.utxo</i> , <i>hsbc.utxo</i>



Unified Ledger

Participant

Tokenized Deposits

Tokenized Government

Bonds

RWA Assets

Traditional Identification

Chain Address

CUSIP + ISIN + Contract

Address

Fragmented identification

systems

.utxo Unified Naming

us-tbond.utxo uk-gilt.utxo

office-tower-hk.utxo

7.3 Central Bank-Level Functionality: PRN Penetrative Regulatory

Nodes

The integration of .utxo with the BIS Unified Ledger provides central banks with regulatory capabilities unattainable in the traditional financial system

【1#L19-L20】:

Core Functions of PRN Nodes (Penetrative Regulatory Nodes):

Function	Description
Real-time Compliance	AML/CFT checks completed before transaction
Verification	execution; vLEI legal entity identity verification
Programmable	Compliance rules automatically executed via smart contracts, no manual intervention required
Regulatory	
Compliance	
Penetrative Auditing	Regulators can access all transaction ledgers in real-time, with commercial privacy protected via MPC

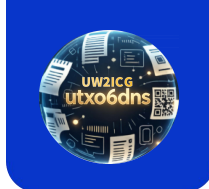


Function	Description
Distributed Key Management	2-of-3 MPC threshold signatures; regulators hold key shards
Audit Attestation	PRNAUDIT RR records immutable audit summaries

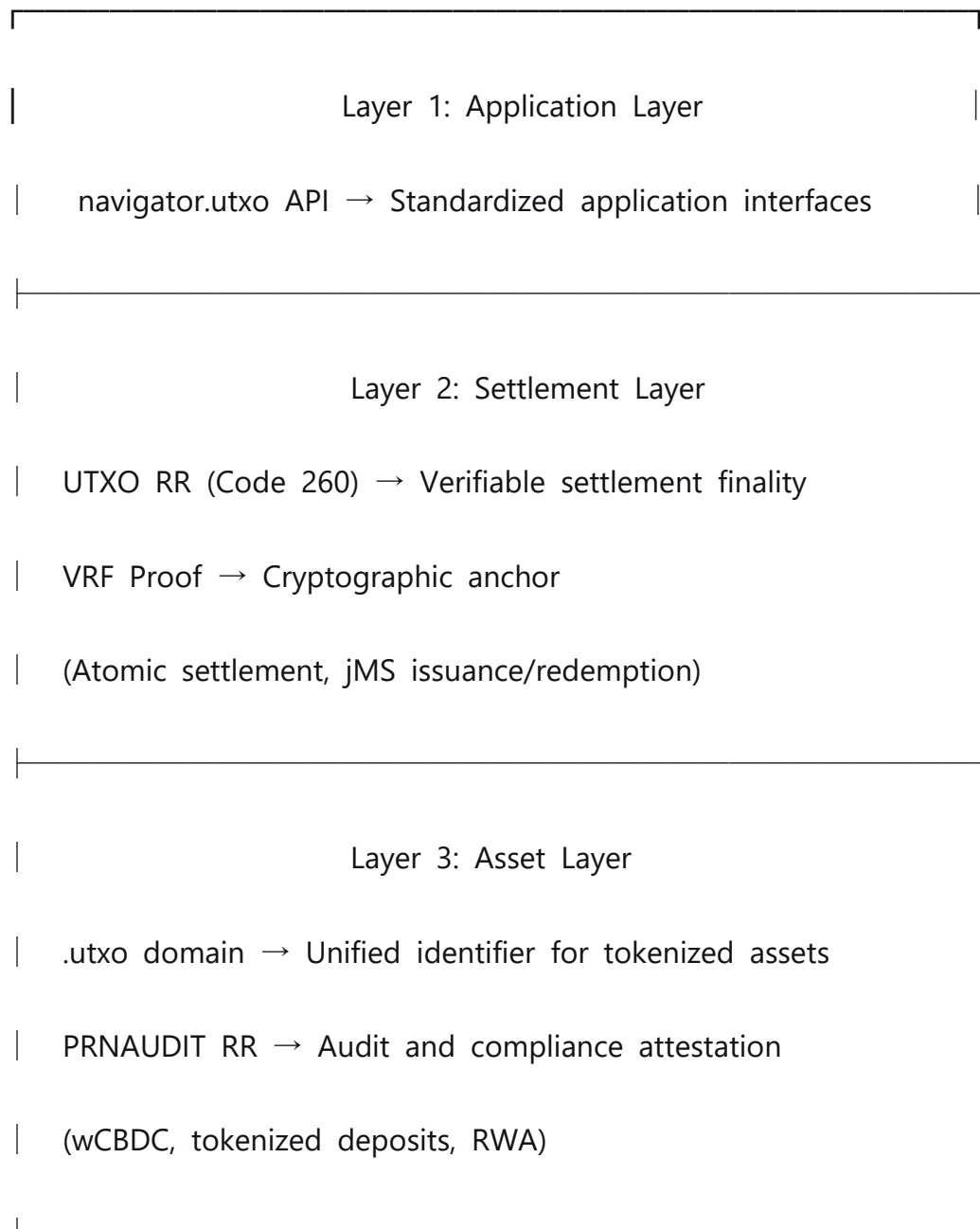
7.4 vLEI: The Trust Anchor for Global Legal Entity Identity

GLEIF's vLEI (Verifiable Legal Entity Identifier) provides legal entity identity verification for the unified ledger 【1†L21】 :

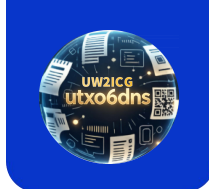
Function	Description
Real-time Verification	Verify legal entity registration status, jurisdiction, authorized representatives
Cryptographic Binding	Cryptographic binding between vLEI and .utxo domain, ensuring domain owner is a real legal entity
Selective Disclosure	Only necessary attributes disclosed during transaction verification, protecting commercial privacy



7.5 Three-Layer Integration Architecture 【1+L22-L24】



VIII. Conclusion: An Irreversible Historical Process



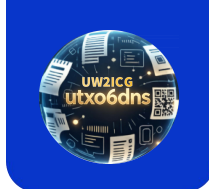
UTXO-DNS is belated — it should have appeared when the HTB proposal was made in 2013. But the 13-year wait has made its arrival timing precisely the critical point at which the global payment system is entering the RWA era 【1+L25】.

8.1 Five Irreversible Drivers 【1+L25】

Driver	Status	Irreversibility
Stablecoin Settlement Network	\$28 trillion annual settlement volume	Already formed global infrastructure parallel to SWIFT
BlackRock RWA Tokenization	\$13 trillion assets on-chain	Institutional endorsement, irreversible
BIS Unified Ledger	Project Agora validation successful	7 central banks + 40+ institutions 实证
Hong Kong Compliance Framework	Stablecoin legislation + licensing	Institutional infrastructure in place
AI Compute and Data Fragmentation	Accelerating	Driven by both market demand and user rights

8.2 .utxo's Strategic Positioning: Triple Naming Layer 【1+L25-L26】

.utxo simultaneously answers three questions of our era:



Era Question

.utxo's Answer

Where is the payment

Unified payment naming layer; any system

endpoint?

addresses via domain names

Whose data is it?

Data UTXO bound to .utxo identity; users have full control

How is compute

Compute RWA uniformly identified and traded

allocated?

via .utxo

The unification of these three naming layers — payment, data, and compute — makes .utxo the most fundamental identifier system for productive factors in the digital age 【1†L26】.

8.3 The Historical Window of Opportunity

What IETF DNSOP is discussing is not merely a new RR type — it is deciding whether DNS is willing to continue serving as the infrastructure for global payments in the RWA era 【1†L27】.

When HTB was overlooked 13 years ago, DNS missed its first opportunity to claim the payment naming layer. In 2026, when the BIS Unified Ledger, \$28 trillion in stablecoin settlement, \$13 trillion in RWA tokenization, and AI compute and data fragmentation all point to the same need — a unified naming layer to identify payment endpoints, data, and compute — DNS can no longer afford to miss this opportunity 【1†L27】.

Facing \$28 trillion in stablecoin settlement and \$13 trillion in RWA tokenization, the goal of UTXO-DNS has never been to change DNS, but to extend DNS — making it the "unified address book" supporting global payments and asset transfers in the RWA era 【1†L27】.

Appendix I: IETF draft-guorong-utxo-dns-01

Core Specification

Document Status: Experimental (Published June 24, 2026)

Author: GT. GuoRong Tian

Co-founders: Z. ZhibinLei, X. XH. Huang, Y. YidanZhu, F. FanYanYang, G. Guo Sheng

Core Specifications 【1†L27】:

- New DNS RR Type: UTXO RR (Code 260)
- EDNS0 Option: Capability Discovery



- Security Mechanism: VRF (RFC 9381 ECVRF-EDWARDS25519-SHA512)
- IPv6 Integration: Generating Verifiable IID, updating RFC 7217
- Compliance Extension: PRN Nodes, including PRNAUDIT RR
- Identity Integration: GLEIF vLEI

Appendix II: Stablecoin and RWA Tokenization

Global Market Data 【1+L28】

Metric	Data	Source
USDC Circulation (end of 2025)	\$75.3 billion	Circle Transparency Report
Stablecoin Annual Settlement	\$28 trillion	Visa/On-chain Data
BlackRock AUM	\$13.46 trillion	BlackRock Q3 2025 Earnings
BUIDL Scale	\$2.8 billion	BlackRock Tokenized Fund
RWA Tokenization Forecast (2030)	\$13-16 trillion	BlackRock/Boston Consulting Group

"Payment, data, and compute — the unification of these three naming layers makes *.utxo* not merely a patch for DNS, but a universal identifier system for the factors of production in the digital age." 【1+L28】